IOQM · GRADE 9 MATHEMATICS

Divisibility, GCD and LCM

Structure, Euclid, reconstruction and prime-product divisibility

TARGET → DIVISOR / MULTIPLE → REDUCTION → THEOREM CHECK → VERIFY

Student learning pack - assimilation, recognition, transfer and mastery

Divisibility, GCD and LCM - Assimilation Book

This book is for a learner who can usually calculate HCF/LCM but is not yet reliable at choosing **why** gcd, lcm, subtraction or Euclid should be the first move.

The target is not a longer formula list. It is a decision habit:

`TARGET -> DIVISOR/MULTIPLE -> DIFFERENCE/REDUCTION -> GCD/LCM -> CHECK`

1. RECONNECT - what is already familiar?

You probably know statements such as:

- `12|60`;
- $gcd(84, 126) = 42$;
- $lcm(12, 18) = 36$;
- digit tests for divisibility by 2, 3, 5, 9 or 11.

Those are useful, but an olympiad problem often hides the important divisibility relation inside words or algebra.

First diagnostic - do not calculate fully

Write only the first thought.

1. Greatest integer that gives the same remainder when dividing 173 and 239.
2. Least positive integer divisible by 12, 15 and 18.
3. Compute $gcd(987, 610)$ efficiently.
4. A number leaves remainder 7 on division by 12, 18 and 30; find the least possible number above 7.
5. $gcd(a, b) = 6$, $lcm(a, b) = 180$; the question asks only for ab.

The intended starts are: **difference**, **lcm**, **Euclid**, **subtract 7 then lcm**, **product invariant**.

If one of these starts felt surprising, that is the bridge this book builds.

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2. DISCOVER - divisibility is an equation, not a symbol trick

$a \mid b$ means:

there is an integer k such that $b = ak$.

This translation is the source of nearly every structural move in this topic.

If $d \mid A$ and $d \mid B$, write $A = dm$, $B = dn$. Then for any integers `r,s`,

$rA + sB = d(rm + sn)$.

So d divides every integer linear combination of A and B.

Why this is stronger than a divisibility test

A divisibility test answers a local question such as “is 7425 divisible by 9?” It reads the digits of one number.

Structural divisibility answers a different kind of question: “what must divide both 4n+7 and 7n+13?” There is no useful digit test because n is unknown. Instead, combine the expressions:

$7(4n + 7) - 4(7n + 13) = -3$.

Any common divisor of the two expressions must divide 3.

Contrast 1 - test or structure?

- 7425 divisible by 9? -> a digit-sum test is cheap.

- $d \mid (4n+7)$ and $d \mid (7n+13)$ -> take a linear combination; a digit test has no role.

Decision question: is the task about one displayed integer, or about a relation that must survive for variable expressions?

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3. MAKE SENSE - subtraction preserves common divisors

Suppose d divides both a and b. Then it divides a-b. Conversely, if $d \mid b$ and $d \mid (a-b)$, then $d \mid a$.

Therefore the common divisors of (a,b) are exactly the common divisors of (b,a-b), so

$gcd(a,b) = gcd(b,a-b)$.

More generally, for any integer q,

$gcd(a,b) = gcd(b,a-qb)$.

This is the invariant behind the Euclidean algorithm.

Euclid as repeated compression

For $gcd(987,610)$:

$987 = 1 \cdot 610 + 377$

so

$gcd(987,610) = gcd(610,377)$.

Continue:

$610 = 1 \cdot 377 + 233$

$377 = 1 \cdot 233 + 144$

$233 = 1 \cdot 144 + 89$

and so on. The pair gets smaller while the common-divisor set stays unchanged.

Contrast 2 - Euclid or factor everything?

- If the numbers factor instantly, factorization may be fine.
- If the pair is large but division gives small remainders, Euclid is usually the cheaper route.

The question is not “which method is legal?” Both may be legal. The question is “which method exposes the invariant with the least work?”

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4. MAKE SENSE - same remainder means differences

If a and b leave the same remainder r when divided by d, then

$a = dq + r$, $b = dp + r$.

Subtract:

$a - b = d(q-p)$.

Therefore

$d \mid (a-b)$.

For several numbers, the common divisor must divide all relevant differences.

Example - unknown divisor

Find the greatest integer that leaves the same remainder when dividing 173 and 239.

The divisor must divide

$239 - 173 = 66.$

The greatest possible divisor is 66. Check:

$173 = 2 \cdot 66 + 41$, $239 = 3 \cdot 66 + 41$.

Example - three numbers

Find the greatest integer giving the same remainder on 221, 323 and 425.

Adjacent differences are both 102, so the greatest divisor is

$gcd(102, 102) = 102.$

Contrast 3 - gcd of numbers or gcd of differences?

- “divides all the numbers” -> gcd of the numbers.
- “leaves the same remainder on all the numbers” -> gcd of differences.

The visible numbers are the same kind of objects, but the words change the first line.

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5. DISCOVER - prescribed remainder is a different problem

Now reverse the unknown.

Suppose **the number `N` is unknown**, but the remainder `r` is prescribed when `N` is divided by several divisors.

If `N` leaves remainder `r` on division by 12, 18 and 30, then

`N-r`

is divisible by all three. So `N-r` is a common multiple of 12, 18 and 30.

For the least such $N > r$,

$N = r + lcm(12, 18, 30).$

For $r = 7$, this gives

$N = 7 + 180 = 187.$

Contrast 4 - the crucial same-remainder fork

- **Unknown divisor, equal remainders from given numbers** -> subtract the given numbers; use gcd of differences.
- **Unknown number, prescribed remainder for given divisors** -> subtract the remainder from the unknown; use a common multiple/lcm.

Do not use the phrase “same remainder” alone as a method label. Ask **what is unknown?**

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6. MAKE SENSE - LCM is the least synchronization point

A common multiple is reached by every given divisor. The lcm is the smallest positive one.

Typical clues:

- least positive integer divisible by several numbers;
- first time several cycles coincide again;
- smallest number satisfying several “is a multiple of” conditions;
- prescribed-remainder reconstruction after subtracting the remainder.

Example - synchronization

Two alarms ring every 12 and 18 minutes. Starting together, they next ring together after

$lcm(12, 18) = 36$

minutes.

Contrast 5 - largest step or first common time?

- greatest step size that fits several distances exactly -> gcd;
- first time several cycles meet -> lcm.

The words “greatest” and “least” are not enough by themselves. Determine whether the object sought is a **divisor** or a **multiple**.

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7. MAKE SENSE - gcd and lcm carry reconstruction information

For positive integers a,b,

$gcd(a, b) \cdot lcm(a, b) = ab$.

A useful normalization is

$a = gu$, $b = gv$, where $g = gcd(a, b)$ and $gcd(u, v) = 1$.

Then the lcm is guv, so if $L = lcm(a, b)$,

uv=L/g.

This separates a shared gcd from a coprime core.

If only the product is requested

If $gcd(a, b) = 6$ and $lcm(a, b) = 180$, then immediately

$ab = 6 \cdot 180 = 1080$.

Do not reconstruct a and b if the question never asks for them.

If the pair is requested

If $g = 12$, $L = 420$, then

uv=L/g=35 and $gcd(u, v) = 1$.

The unordered coprime factor splits of 35 are (1,35) and (5,7), giving

$(a, b) = (12, 420)$ or (60,84) up to order.

This is **reconstruction**, not just the product identity.

Contrast 6 - invariant or reconstruction?

- target ab -> stop at $ab = gL$;
- target the actual pair(s) -> normalize, use uv=L/g, and enforce coprimality.

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8. MAKE SENSE - divisibility chains compress several conditions

If

$a \mid b$ and $b \mid c$,

then $a \mid c$ because $b = am$ and $c = bn$ imply $c = a(mn)$.

This transitivity gives immediate gcd/lcm facts:

If $a \mid b$, then
$gcd(a, b) = a$, $lcm(a, b) = b$.
If $a \mid b \mid c$, then
$gcd(a, b, c) = a$, $lcm(a, b, c) = c$.

Example - an unknown inside a chain

Suppose 6**|**x and $x \mid 72$.
Do not treat these as unrelated tests. Write
$6 \mid x \mid 72$.
So x must be a divisor of 72 that is also a multiple of 6. The chain is the organizing representation.

Contrast 7 - chain or separate checking?

When one divisibility relation feeds another, transitivity may remove whole branches of work. Independent checking is useful only after the chain has been compressed.

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9. TRY - attempt before help

For each problem, make a first attempt **before** reading any support. If stuck, use the fullest support first; on later problems, work with progressively less support until you can solve independently.

Problem A

Find the greatest integer that leaves the same remainder when dividing 437, 581 and 725.
Full support: compute the differences $581 - 437$ and $725 - 581$, then take their gcd.
Structural prompt: an unknown common divisor must divide every difference.
Recognition prompt: equal remainders disappear under subtraction.
Independent target: later problems give no support.
Answer: 144.

Problem B

Find the least $N > 9$ that leaves remainder 9 when divided by 12, 15 and 20.
Full support: $N - 9 = lcm(12, 15, 20)$ for the least solution.
Structural prompt: remove the prescribed remainder so that every divisor divides the same number.
Recognition prompt: the unknown is the number being constructed, not the divisor.
Independent target: later problems give no support.
Answer: 69.

Problem C

Compute $gcd(2025, 748)$.
Full support: repeatedly use $a = qb + r$ and replace `(a,b)` by `(b,r)`.
Structural prompt: preserve the common-divisor set while shrinking the pair.
Recognition prompt: division with remainder is cheaper than full factorization here.
Independent target: later problems give no support.
Answer: 1.

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10. DIAGNOSE - tempting wrong moves

Error A - “same remainder means lcm”

Why tempting: remainder construction often uses lcm.

Repair: ask **which object is unknown?** Unknown divisor -> differences/gcd. Unknown number with prescribed remainder -> subtract remainder/lcm.

Error B - “gcd means factor both numbers”

Why tempting: school examples often teach HCF through prime factors.

Repair: choose a representation. Euclid can preserve the same gcd while making the numbers much smaller.

Error C - “divisibility test solves every divisibility problem”

Why tempting: tests are memorable.

Repair: tests detect a property of one integer. Variable expressions and common-divisor conditions require structural algebra.

Error D - using $gcd \cdot lcm = ab$ without deciding the target

Why tempting: the identity looks powerful.

Repair: if the pair itself is required, the product identity is only the first constraint; normalize and enforce $gcd(u, v) = 1$.

Error E - ignoring a divisibility chain

Why tempting: each condition looks separate.

Repair: write the chain first. Transitivity can make some conditions automatic.

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11. ADOPT - the first-move rules

1. Translate $a \mid b$ into $b = ak$ when a proof or variable relation is hidden.
2. For a common divisor of expressions, take integer linear combinations.
3. For equal remainders with unknown divisor, subtract the given numbers.
4. For large gcd pairs, try Euclidean reduction before full factorization.
5. For least simultaneous divisibility, build an lcm.
6. For a prescribed remainder, subtract the remainder before building the lcm.
7. If gcd and lcm are both given, use $gL = ab$; reconstruct only if necessary.
8. Compress $a \mid b \mid c$ as a chain before enumerating cases.

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12. VALIDATED IOQM ANCHORS - source to mechanism

IOQM-2025-Q02

The official item is a counting question about integers up to 100 divisible by 3 but not by 2. The structural move is to count multiples of 3 and remove those also divisible by 2, i.e. multiples of 6. The independently checked answer is 17.

Learning use: divisibility condition -> common-multiple overlap. This is not a claim that inclusion-exclusion belongs canonically to this topic; the counting machinery is a bridge.

IOQM-2025-Q27

The official item asks for ordered positive integer triples under an equation involving two lcm terms. The first decisive conversion is

`lcm(x,c)=xc/gcd(x,c).`

This turns the lcm equation into restrictions on gcd values and collapses the search to two symmetric families. The independently checked answer is 40.

Learning use: when lcm appears inside an algebraic relation, rewrite it through gcd structure before brute-force enumeration.

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13. TRANSFER - same invariant, changed surface

Representation change

"Several measurements have the same remainder when measured in units of length d." Translate to differences divisible by d.

Context change

"Three machines reset every 18, 24 and 40 minutes." Translate to the first positive common multiple.

Spacing change

"Marks are at positions 1001, 1457 and 1913 on a line. What is the largest equal step that lands on all marks after a common offset?" Translate to gcd of differences.

Downstream modular bridge

A later modular-arithmetic chapter may write equal-remainder information with congruence notation. The underlying fact from this chapter remains $d \mid (a-b)$; the later chapter retrieves it rather than requiring this chapter to introduce its notation.

Final self-check

Before calculating, can you state:

- the target object: divisor or multiple?
- the smallest useful representation?
- the first useful line?
- the nearby tempting route and why it is less direct?
- a final divisibility/remainder/coprimality check?

If yes, you are using the topic structurally rather than procedurally.

NT-01 - First-Step Reference

Use this after the Assimilation Book. It is a compression layer, not a replacement for the reasoning.

Recognition atlas

Visible clue	Structural question	First useful move
$a \mid b$, common divisor of algebraic expressions	what integer combination removes variables/terms?	rewrite as integer multiples; combine
prime p and $p \mid ab$	is the divisor really prime?	apply Euclid's Lemma: $p \mid a$ or $p \mid b$
greatest integer dividing all given numbers	common divisor target?	gcd of the numbers
same remainder on several given numbers; divisor unknown	what disappears under subtraction?	take differences; gcd them if greatest requested
least number with a prescribed remainder under several divisors	can I remove the remainder first?	`N-r` is a common multiple
least positive number divisible by several integers	multiple target?	lcm
first time cycles coincide	synchronization?	lcm
largest common step/spacing	divisor of all distances?	gcd
large gcd pair	can remainder division shrink it?	Euclidean algorithm
gcd and lcm both given	only product or actual pair?	$gL = ab$; normalize if reconstructing
$a \mid b \mid c$-style nested constraints	is there a chain?	use transitivity before casework
one explicit integer and a familiar digit pattern	local yes/no check only?	divisibility test may be cheapest

Decision router

```
1. What is unknown?
   |
   +-- a DIVISOR -> common divisor? same-remainder differences? gcd?
   |
   +-- a NUMBER/MULTIPLE -> least common multiple? remove prescribed remainder first?

2. Is a PRIME divisor known to divide a product?
   -> check primality, then Euclid's Lemma: p|ab => p|a or p|b

3. Is gcd computation itself large?
   -> Euclid: (a,b) -> (b, a mod b)

4. Are gcd and lcm both given?
   -> gL=ab
   -> if actual a,b needed: a=gu, b=gv, gcd(u,v)=1, uv=L/g

5. Are conditions nested?
   -> compress divisibility chain before enumeration
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First-step cards

Divisibility algebra
$d \mid A$ and $d \mid B$ -> $d \mid (rA + sB)$ for integers r, s.

Euclid's Lemma - prime divisor of a product
If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

A short proof uses the same gcd structure as this topic. If p does not divide a, then $gcd(p,a) = 1$. Therefore integers r, s exist with $rp + sa = 1$. Multiply by b:

$rpb + sab = b$.

Both terms on the left are divisible by p, so $p \mid b$.

Hypothesis check: primality matters. The composite imitation is false: $6 \mid 2 \cdot 3$, but 6 divides neither 2 nor 3.

Decision boundary: use the lemma to split a **prime** divisor across a product. Do not use it for an arbitrary composite divisor, and do not replace a shorter direct divisibility argument with theorem-name recall.

Same remainder, unknown divisor
$a = dq + r$, $b = dp + r$ -> $d \mid (a - b)$.

Prescribed remainder, unknown number
N leaves remainder r under each divisor -> every divisor divides N-r.

Euclid
$a = qb + r$ -> $gcd(a,b) = gcd(b,r)$.

gcd/lcm reconstruction
$a = gu$, $b = gv$, $gcd(u,v) = 1$, uv=L/g.

Divisibility chain
$a \mid b$ and $b \mid c$ -> $a \mid c$.

Contrast strip - why the first move changes

1. **gcd vs lcm:** divisor target -> gcd; common-multiple target -> lcm.
2. **same remainder fork:** unknown divisor -> differences; unknown number with prescribed remainder -> subtract remainder then lcm.
3. **prime divisor vs composite divisor:** $p \mid ab$ with prime p -> Euclid's Lemma; a composite divisor need not divide either factor.
4. **test vs structure:** one explicit integer -> divisibility test may work; variable/common-divisor relation -> translate/combine algebraically.
5. **Euclid vs factoring:** shrink by remainders when full factorization is expensive.
6. **numbers vs differences:** common divisor of the numbers -> gcd numbers; equal remainders -> gcd differences.
7. **product vs pair:** $gL = ab$ answers a product question; pair reconstruction also needs coprime normalization.
8. **chain vs independent checks:** $a \mid b \mid c$ contains transitive information; use it first.
9. **largest step vs first meeting:** spacing asks for gcd; synchronization asks for lcm.

30-second checks

Before committing to a method:

- What is the target object?
- If I want to split divisibility across a product, is the divisor prime?
- What disappears if I subtract?
- Can I remove a prescribed remainder?
- Is Euclid cheaper than factoring?
- Am I using an identity beyond what it actually determines?
- Is there a divisibility chain?
- Can I verify the final remainder/divisibility/coprimality condition directly?

NT-01 - Recognition and First-Line Lab

Do not fully solve unless the prompt asks. Write only the **first useful mathematical line** and, where requested, one short reason.

1. Greatest integer giving the same remainder on 188 and 296.
2. Least positive integer divisible by 14, 20 and 35.
3. Compute $gcd(12345, 5432)$ efficiently.
4. $gcd(a, b) = 9$, $lcm(a, b) = 252$; write the invariant involving ab.
5. A divisor d divides both 5x+7y and 2x+3y; write a useful linear combination.
6. A number `N` leaves remainder 4 on division by 6, 10 and 15; write the first line for the least $N > 4$.
7. Greatest divisor giving the same remainder on 812, 1046 and 1280; write only the reduction.
8. Two alarms ring every 18 and 24 minutes; write only the operation for the next common ringing time.
9. $a = gu$, $b = gv$, where $g = gcd(a, b)$; state the required condition on `u,v`.
10. Decide whether `84|1260` by writing an integer-quotient statement.
11. $d \mid (4n + 7)$ and $d \mid (7n + 13)$; write a combination independent of n.
12. `6|x` and $x \mid 72$; rewrite the information as one chain.
13. Three points lie at 1001, 1457 and 1913 on a number line. A common step size must fit all pairwise offsets. Write the first reduction.
14. The question asks only for ab when $gcd(a, b) = 12$, $lcm(a, b) = 420$. Write the first line and stop.
15. The question asks for all possible pairs when $gcd(a, b) = 12$, $lcm(a, b) = 420$. Write the normalization after the product identity.
16. A student begins prime-factorising two large numbers to find their gcd, but the first division gives a small remainder. Write the alternative first move.

Recognition-only pass

For each item, identify the target type without solving: `DIVISOR`, `MULTIPLE`, `REDUCTION`, `RECONSTRUCTION`, `CHAIN`, or `LOCAL_TEST`.

The later independent mastery check removes these labels.

NT-01 - Practice and Transfer Bank

All items here are author-created. Historical IOQM IDs are not assigned to them.

F0 - Foundation repair

1. State whether 18|234 and justify with an integer quotient.
2. If $d \mid 48$ and $d \mid 30$, show directly that $d \mid 18$.
3. Compute $gcd(84, 126)$ and $lcm(84, 126)$.
4. If 8|24 and 24|120, state two consequences about divisibility and gcd/lcm.

F1 - Direct recognition

5. Find the greatest integer giving the same remainder on 155 and 239.
6. Find the least positive integer divisible by 16, 20 and 24.
7. Use Euclid to compute $gcd(2025, 748)$.
8. Find the least $N > 7$ leaving remainder 7 on division by 12, 18 and 30.
9. If $gcd(a, b) = 12$ and $lcm(a, b) = 420$, determine ab.

F2 - Standard structural use

10. Find the greatest integer giving the same remainder on 437, 581 and 725.
11. Find all positive integers d such that 305, 473 and 641 leave the same remainder on division by d.
12. Find the least $N > 9$ leaving remainder 9 on division by 12, 15 and 20.
13. Positive integers a,b have gcd 12 and lcm 420. List the possible unordered pairs (a,b).
14. Find all positive integers x satisfying 6|x and $x \mid 72$.

F3 - Disguised structure

15. If d divides both 7x+5y and 3x+2y, prove that $d \mid x$ and $d \mid y$. Hence show that no $d > 1$ is possible when $gcd(x, y) = 1$.
16. Three machines reset every 18, 24 and 40 minutes. They reset together at 09:00. When do they next reset together?
17. Suppose $d \mid (4n + 7)$ and $d \mid (7n + 13)$. Determine all positive values d that can occur for some integer n.
18. Positive integers a,b have gcd 15 and lcm 900. Write $a = 15u$, $b = 15v$ and list the possible unordered pairs (a,b).
19. A ruler has marks at 1001, 1457 and 1913 mm. What is the largest positive step size d for which all three positions have the same remainder when divided by d?

F4 - Preliminary-style transfer

20. Find the greatest d such that 1001, 1457 and 1913 leave the same remainder upon division by d.
21. Find the least positive integer N such that N+5 is divisible by 18, 24 and 30.
22. A positive integer d divides both 11n+8 and 7n+5. Find the largest possible value of d over all integers n.
23. Positive integers $a, b, c \leq 50$ satisfy $27(lcm(a, c) + lcm(b, c)) = 26c(a + b)$. Let $x = gcd(a, c)$ and $y = gcd(b, c)$. Re-derive the key restriction on x,y before any enumeration. Do not use the historical answer as a starting assumption.
24. A number N leaves remainder 5 when divided by 12 and 18, but remainder 0 when divided by 5. Find the least $N > 5$ satisfying all three conditions, or prove none exists.

Transfer prompts

25. **Representation change:** express “a and b leave the same remainder when divided by d” using only divisibility language, without congruence notation.

26. **Context change:** explain why a largest equal-spacing problem and a first-synchronization problem usually route to different operations.
27. **Changed target:** two numbers have gcd 18 and lcm 630. Explain why the product is determined but the ordered pair need not be unique.
28. **Downstream bridge:** state exactly which divisibility fact a later modular-arithmetic chapter may retrieve when it writes a and b as having the same residue modulo d.
29. **WHY-NOT:** Why is lcm the wrong first move for “greatest divisor giving the same remainder on 437, 581 and 725”?
30. **WHY-NOT:** Why do $gcd(a,b) = 12$, $lcm(a,b) = 420$ not force only the pair `(12,420)`?

NT-01 - Independent Mastery Check

No default hints. No method labels are attached to the questions. Show enough reasoning to justify the first move and the final check.

Part A - Notice and first useful line

1. Greatest positive integer giving the same remainder on 428, 596 and 764. Write only the first reduction.
2. Least positive integer greater than 11 leaving remainder 11 when divided by 18, 24 and 30. Write only the first equation.
3. $d \mid (13n+9)$ and $d \mid (8n+5)$. Write one integer linear combination that sharply restricts d.
4. $gcd(a,b) = 14$, $lcm(a,b) = 840$; write the first invariant if the target is ab.

Part B - Full mixed solve

5. Compute $gcd(123456, 7890)$ without completely prime-factorising either number.
6. Find the greatest positive integer giving the same remainder on 428, 596 and 764.
7. Find the least positive integer greater than 11 leaving remainder 11 on division by 18, 24 and 30.
8. Positive integers a, b satisfy $gcd(a,b) = 14$, $lcm(a,b) = 840$. Find ab.
9. A positive integer d divides both 11n+8 and 7n+5. Find the largest possible d for some integer n.
10. Find all positive integers d such that 305, 473 and 641 leave the same remainder on division by d.
11. The lcm of two positive integers is 360 and their gcd is 12. If one number is 72, find the other and verify both gcd and lcm.
12. Three events recur every 28, 42 and 63 days and occur together today. After how many days do they next occur together?

Part C - Same surface, different decision

13. Problem A: find the greatest integer that gives the same remainder when dividing 221, 323 and 425.
    Problem B: find the least integer $N > 5$ that leaves remainder 5 on division by 6, 10 and 15.
    Solve both and state in one sentence why their first moves differ.

14. Problem A: determine whether 7425 is divisible by 9.
    Problem B: if $d \mid (4n+7)$ and $d \mid (7n+13)$, determine the possible positive d.
    Solve both and state why the first problem may use a divisibility test while the second should use structural divisibility.

Part D - Transfer and WHY-NOT

15. Marks are placed at positions 812 m, 1046 m and 1280 m along a route. Find the largest unit length d for which all three positions have the same offset from a unit boundary. State the hidden arithmetic model.
16. A learner says: "Because same remainders are mentioned, I always take an lcm." Give a counterexample from this test and write the correct decision boundary in one sentence.

Mastery reflection

For any three solved items, add:

- What visible clue mattered?
- What was the hidden structure?
- What was your first useful line?
- What nearby method was tempting but less direct or wrong?
- What final check did you perform?